

블록 재구성을 통한 training 후 양자화의 한계 극복

BRECQ: Pushing the Limit of Post-Training Quantization by Block Reconstruction

김동희

Accepted at ICLR 2021

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01. Introduction

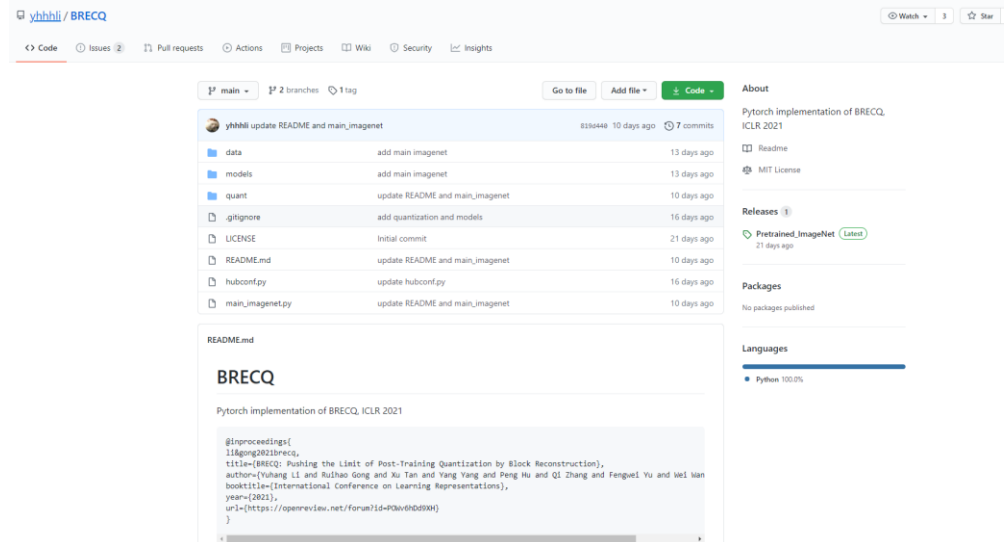
Why select this paper?

Rating: 7: Good paper, accept

8: Top 50% of accepted papers, clear accept

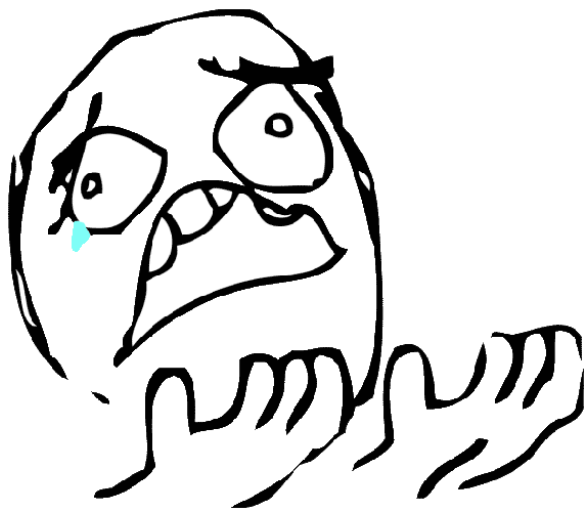
6: Marginally above acceptance threshold

7: Good paper, accept



01. Introduction

Why select this paper?



Proof. We will prove the theorem using quadratic form. Denote the weight vector shape as $\theta \in \mathbb{R}^d$, and the network output vector shape as $\mathbf{z}^{(n)} \in \mathbb{R}^m$. The quadratic form of the $\Delta\theta^\top \mathbf{H}^{(\theta)} \Delta\theta$ can be represented by:

$$\Delta\theta^\top \mathbf{H}^{(\theta)} \Delta\theta = \sum_{i=1}^d \Delta\theta_i^2 \left(\frac{\partial^2 L}{\partial \theta_i^2} \right) + 2 \sum_{i < j}^d \Delta\theta_i \Delta\theta_j \frac{\partial L}{\partial \theta_i \theta_j} = \sum_{i=1}^d \sum_{j=1}^d \left(\Delta\theta_i \Delta\theta_j \frac{\partial L}{\partial \theta_i \theta_j} \right), \quad (12)$$

where L is the cross-entropy loss. Based on Eq. (5), we have

$$\frac{\partial^2 L}{\partial \theta_i \theta_j} = \sum_{k,l}^m \frac{\partial \mathbf{z}_k^{(n)}}{\partial \theta_i} \frac{\partial^2 L}{\partial \mathbf{z}_k^{(n)} \mathbf{z}_l^{(n)}} \frac{\partial \mathbf{z}_l^{(n)}}{\partial \theta_j} \quad (13)$$

Substituting above equation in Eq. (12), we have

$$\Delta\theta^\top \mathbf{H}^{(\theta)} \Delta\theta = \sum_{i=1}^d \sum_{j=1}^d \Delta\theta_i \Delta\theta_j \left(\sum_{k=1}^m \sum_{l=1}^m \frac{\partial \mathbf{z}_k^{(n)}}{\partial \theta_i} \frac{\partial^2 L}{\partial \mathbf{z}_k^{(n)} \mathbf{z}_l^{(n)}} \frac{\partial \mathbf{z}_l^{(n)}}{\partial \theta_j} \right) \quad (14a)$$

$$= \sum_{i=1}^d \sum_{j=1}^d \sum_{k=1}^m \sum_{l=1}^m \left(\Delta\theta_i \Delta\theta_j \frac{\partial \mathbf{z}_k^{(n)}}{\partial \theta_i} \frac{\partial^2 L}{\partial \mathbf{z}_k^{(n)} \mathbf{z}_l^{(n)}} \frac{\partial \mathbf{z}_l^{(n)}}{\partial \theta_j} \right) \quad (14b)$$

$$= \sum_{k=1}^m \sum_{l=1}^m \left(\frac{\partial^2 L}{\partial \mathbf{z}_k^{(n)} \mathbf{z}_l^{(n)}} \right) \left(\sum_{i=1}^d \Delta\theta_i \frac{\partial \mathbf{z}_k^{(n)}}{\partial \theta_i} \right) \left(\sum_{j=1}^d \Delta\theta_j \frac{\partial \mathbf{z}_l^{(n)}}{\partial \theta_j} \right) \quad (14c)$$

$$= (\Delta\theta \mathbf{J} \left[\frac{\mathbf{z}^{(n)}}{\theta} \right])^\top \mathbf{H}^{(\mathbf{z}^{(n)})} (\Delta\theta \mathbf{J} \left[\frac{\mathbf{z}^{(n)}}{\theta} \right]), \quad (14d)$$

where we define the $\mathbf{J} \left[\frac{x}{y} \right]$ is the Jacobian matrix of x w.r.t. y . To this end, we use the first-order Taylor expansion as we did in Eq. (8) to approximate the change in network output, i.e.,

$$\Delta \mathbf{z}^{(n)} \approx \Delta\theta \mathbf{J} \left[\frac{\mathbf{z}^{(n)}}{\theta} \right] \quad (15)$$

Therefore, the final objective is transformed to $\Delta \mathbf{z}^{(n),\top} \mathbf{H}^{(\mathbf{z}^{(n)})} \Delta \mathbf{z}^{(n)}$. $\square$

01. Introduction

Why select this paper?

General Response

ICLR 2021 Conference Paper1 Authors

16 Nov 2020 ICLR 2021 Conference Paper1 Official Comment Readers:  Everyone

Comment:

We feel grateful for the reviewers' positive feedback towards BRECQ. Some of them raise the concern that the method is not clear. We would like to make some rough clarifications about this point.

BRECQ studies the loss degradation in post-training quantization by approximating the Taylor expansion and Gauss-Newton approximation of the Hessian matrix. We show that the cross-layer dependence can be computed by measuring the distance of the output of the whole network. However, if more layers are considered, the distance of output will become an inaccurate signal and prohibits the optimization. To reach the best tradeoff between them, we use the building block as the basic reconstruction unit. This method requires no hyper-parameters and is simple to perform. We conduct extensive experiments on various models and tasks to verify the algorithm.

01. Introduction

Presentation Goal

논문엔 있지만 오늘 발표에는 담겨져 있지 않은 내용:

Effect of the first layer and the last layer

Calibration data point 수에 따른 quantization 성능 변화

Quantization step size s 를 학습시키는 방법

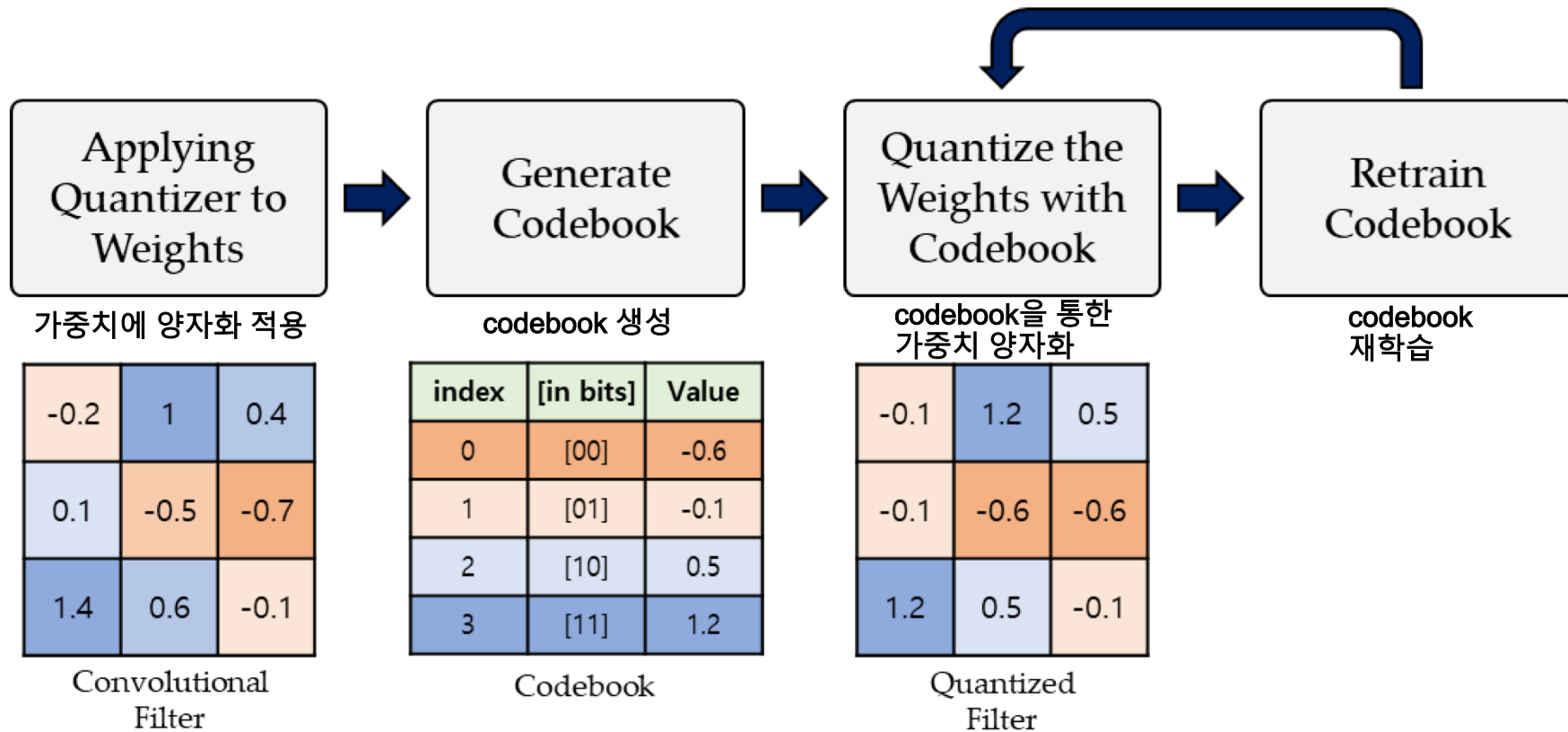
Mixed precision을 학습시키기 위한 genetic algorithm

이외 수식 & 증명.....

Quantization을 위한 논문의 핵심 접근 방법에 집중!

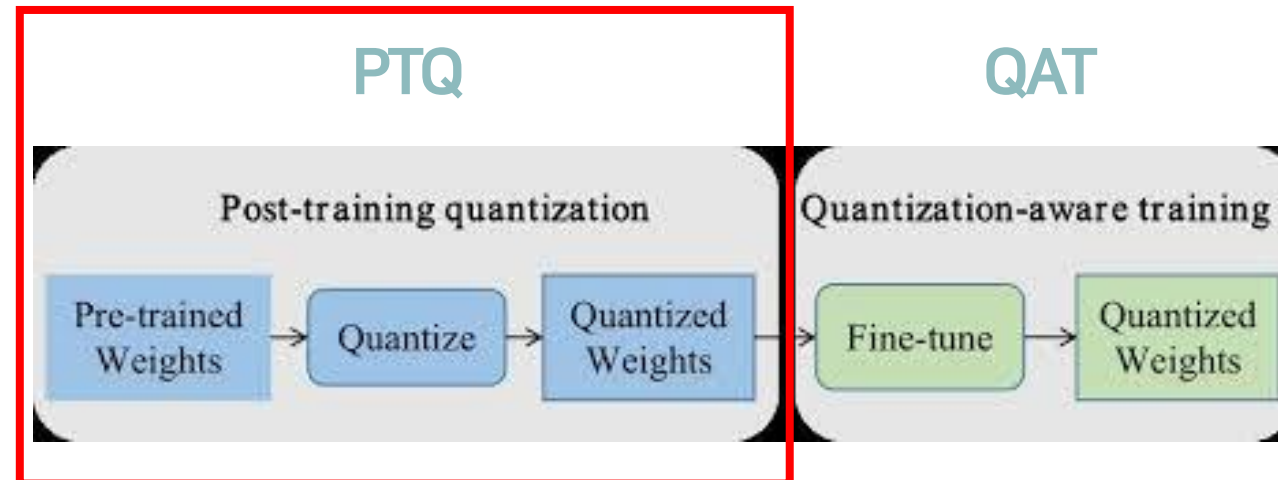
01. Introduction

Overview



01. Introduction

Overview



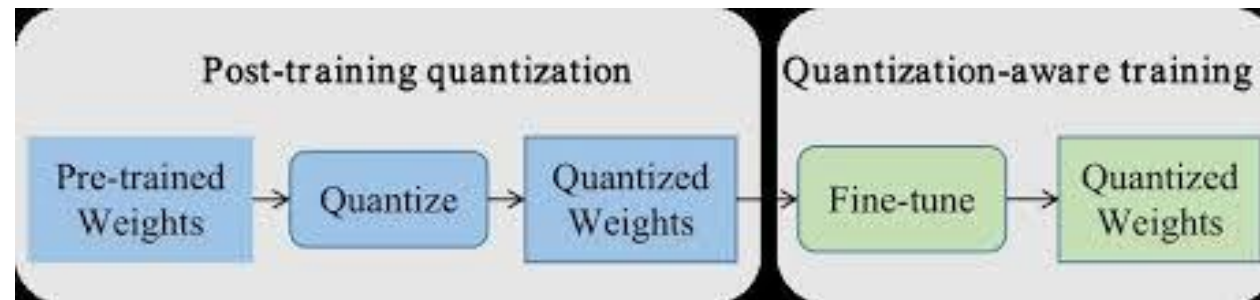
X240 faster production!

01. Introduction

Overview

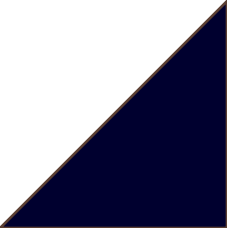
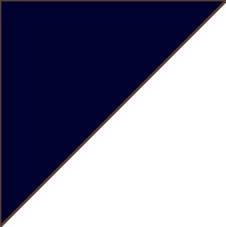
PTQ

QAT



($z = Wx$), and the quantized model can be obtained by a layer-by-layer feature map reconstruction algorithm (with few calibration images). Recent works, like Bit-Split (Wang et al., 2020) and

그렇다고 데이터가 하나도 안 필요한 건 아니고...



Q&A

02. Background

Loss Degradation

BRECQ studies the **loss degradation** in post-training quantization by approximating the Taylor expansion and Gauss-Newton approximation of the Hessian matrix.

$$\min ||\hat{\mathbf{w}} - \mathbf{w}||_F^2. \text{ s.t. } \hat{\mathbf{w}} \in \mathcal{Q}_b^{\text{u,sym}}$$

Quantized weight와 Original weight 간의 distance 를 최소화

=> 2018년까지는 좋은 접근방법

$$\mathcal{Q}_b^{\text{u,sym}} = s \times \{-2^{b-1}, \dots, 0, \dots, 2^{b-1} - 1\}$$

02. Background

Loss Degradation

BRECQ studies the **loss degradation** in post-training quantization by approximating the **Taylor expansion** and **Gauss-Newton approximation** of the **Hessian matrix**.

$$\mathcal{Q}_b^{\text{u,sym}} = s \times \{-2^{b-1}, \dots, 0, \dots, 2^{b-1} - 1\}$$

$$\min \mathbb{E}[L(\hat{\mathbf{w}})], \text{ s.t. } \hat{\mathbf{w}} \in \mathcal{Q}_b^{\text{u,sym}}$$

loss function을 최소화하는 Quantized weight 찾기

02. Background

Taylor Expansion

$$\bar{\mathbf{g}}^{(\mathbf{w})} = \mathbb{E}[\nabla_{\mathbf{w}} L]$$
$$\bar{\mathbf{H}}^{(\mathbf{w})} = \mathbb{E}[\nabla_{\mathbf{w}}^2 L]$$

BRECQ studies the loss degradation in post-training quantization by approximating the **Taylor expansion** and Gauss-Newton **approximation** of the Hessian matrix.

Taylor series

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 \cdots \frac{f^{(n)}(a)}{n!}(x-a)^n = \sum_{n=0}^{\infty} \frac{f^{(n)}(x-a)}{n!}(x-a)^n$$

$$f(x) - f(a) = f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2$$

$$\mathbb{E}[L(\mathbf{w} + \Delta \mathbf{w})] - \mathbb{E}[L(\mathbf{w})] \approx \Delta \mathbf{w}^T \bar{\mathbf{g}}^{(\mathbf{w})} + \frac{1}{2} \Delta \mathbf{w}^T \bar{\mathbf{H}}^{(\mathbf{w})} \Delta \mathbf{w}$$

$$H(f) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

Hessian 의 complexity 는 $O(W^2)$ => Hessian 계산을 위해 두 가지 가정을 진행

02. Background

Hessian Matrix

$$\bar{\mathbf{g}}^{(\mathbf{w})} = \mathbb{E}[\nabla_{\mathbf{w}} L]$$
$$\bar{\mathbf{H}}^{(\mathbf{w})} = \mathbb{E}[\nabla_{\mathbf{w}}^2 L]$$

BRECQ studies the loss degradation in post-training quantization by approximating the Taylor expansion and Gauss-Newton approximation of the **Hessian matrix**.

$$\mathbb{E}[L(\mathbf{w} + \Delta \mathbf{w})] - \mathbb{E}[L(\mathbf{w})] \approx \Delta \mathbf{w}^T \bar{\mathbf{g}}^{(\mathbf{w})} + \frac{1}{2} \Delta \mathbf{w}^T \bar{\mathbf{H}}^{(\mathbf{w})} \Delta \mathbf{w}$$

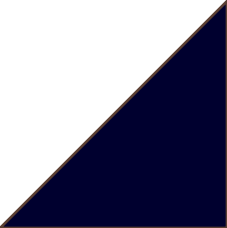
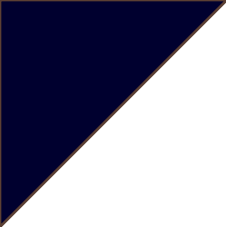
1. Layers are mutual-independent, $\bar{\mathbf{H}}^{(\mathbf{w}^{(\ell)})} = \mathbb{E}[\mathbf{x}^{(\ell)} \mathbf{x}^{(\ell),T} \otimes \mathbf{H}^{(\mathbf{z}^{(\ell)})}]$, where $\otimes$ is the Kronecker product.

2. $\mathbf{H}^{(\mathbf{z}^{(\ell)})} = c \times \mathbf{I}$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \otimes \begin{bmatrix} 0 & 5 \\ 6 & 7 \end{bmatrix} = \begin{bmatrix} 1 \cdot 0 & 1 \cdot 5 & 2 \cdot 0 & 2 \cdot 5 \\ 1 \cdot 6 & 1 \cdot 7 & 2 \cdot 6 & 2 \cdot 7 \\ 3 \cdot 0 & 3 \cdot 5 & 4 \cdot 0 & 4 \cdot 5 \\ 3 \cdot 6 & 3 \cdot 7 & 4 \cdot 6 & 4 \cdot 7 \end{bmatrix} = \begin{bmatrix} 0 & 5 & 0 & 10 \\ 6 & 7 & 12 & 14 \\ 0 & 5 & 0 & 20 \\ 18 & 21 & 24 & 28 \end{bmatrix}$$

전반적으로 좋은 성능을 보이지만 INT2 라는 극한 환경에서는 성능이 떨어짐

=> $\Delta \mathbf{W}$ 가 커질수록 위의 가정이 성립되지 않기 때문(이라고 주장)



Q&A

03. Proposed Method

Recap

Weight를 full precision과 최대한 따라하자



Weight $\hookrightarrow$ loss값을 비슷하게 만드는 quantization 방법을 찾아보자



근데 그 방법 찾아보니까 weight에 대한 Hessian matrix를 구해야 하네?



Hessian 직접 구하기는 너무 힘들니까 대충 가정 깔고 들어가자



$\hookrightarrow$ 대충 가정 깔지 말고 **그가정을 좀더 strict하게 유도해보자!**

(본 논문의 핵심 아이디어)

03. Proposed Method

Cross-Layer Dependency

$$\bar{\mathbf{g}}^{(\mathbf{w})} = \mathbb{E}[\nabla_{\mathbf{w}} L]$$

$$\bar{\mathbf{H}}^{(\mathbf{w})} = \mathbb{E}[\nabla_{\mathbf{w}}^2 L]$$

$$\mathbb{E}[L(\mathbf{w} + \Delta \mathbf{w})] - \mathbb{E}[L(\mathbf{w})] \approx \Delta \mathbf{w}^T \bar{\mathbf{g}}^{(\mathbf{w})} + \frac{1}{2} \Delta \mathbf{w}^T \bar{\mathbf{H}}^{(\mathbf{w})} \Delta \mathbf{w}$$

We show that the **cross-layer dependence** can be computed by measuring the distance of the output of the whole network.

첫 번째 가정

Layers are mutual-independent, $\bar{\mathbf{H}}^{(\mathbf{w}^{(\ell)})} = \mathbb{E}[\mathbf{x}^{(\ell)} \mathbf{x}^{(\ell),T} \otimes \mathbf{H}^{(\mathbf{z}^{(\ell)})}]$, where $\otimes$ is the Kronecker product.

=> Layer dependence를 고려해서 Hessian을 다시 계산해 보자

$$\arg \min_{\hat{\theta}} \Delta \theta^T \bar{\mathbf{H}}^{(\theta)} \Delta \theta \approx \arg \min_{\hat{\theta}} \mathbb{E} \left[\Delta \mathbf{z}^{(n),T} \mathbf{H}^{(\mathbf{z}^{(n)})} \Delta \mathbf{z}^{(n)} \right]$$

$$\theta = \text{vec}[\mathbf{w}^{(1),T}, \dots, \mathbf{w}^{(n),T}]^T$$

증명은 논문 참조...

03. Proposed Method

Approximating Pre-activation Hessian

$$\bar{\mathbf{g}}^{(\mathbf{w})} = \mathbb{E}[\nabla_{\mathbf{w}} L]$$

$$\bar{\mathbf{H}}^{(\mathbf{w})} = \mathbb{E}[\nabla_{\mathbf{w}}^2 L]$$

$$\mathbb{E}[L(\mathbf{w} + \Delta \mathbf{w})] - \mathbb{E}[L(\mathbf{w})] \approx \Delta \mathbf{w}^T \bar{\mathbf{g}}^{(\mathbf{w})} + \frac{1}{2} \Delta \mathbf{w}^T \bar{\mathbf{H}}^{(\mathbf{w})} \Delta \mathbf{w}$$

We show that the **cross-layer dependence** can be computed by measuring the distance of the output of the whole network.

두 번째 가정

$$\mathbf{H}(\mathbf{z}^{(\ell)}) = c \times \mathbf{I}$$

$$\arg \min_{\hat{\theta}} \Delta \theta^T \bar{\mathbf{H}}^{(\theta)} \Delta \theta \approx \arg \min_{\hat{\theta}} \mathbb{E} \left[\Delta \mathbf{z}^{(n),T} \mathbf{H}(\mathbf{z}^{(n)}) \Delta \mathbf{z}^{(n)} \right] \quad \text{이} \quad \|\Delta \mathbf{z}^{(\ell)}\|^2 \quad \text{가 됨}$$

저런 가정은 information의 손실이 많다!

Fisher Information Matrix를 적용하면

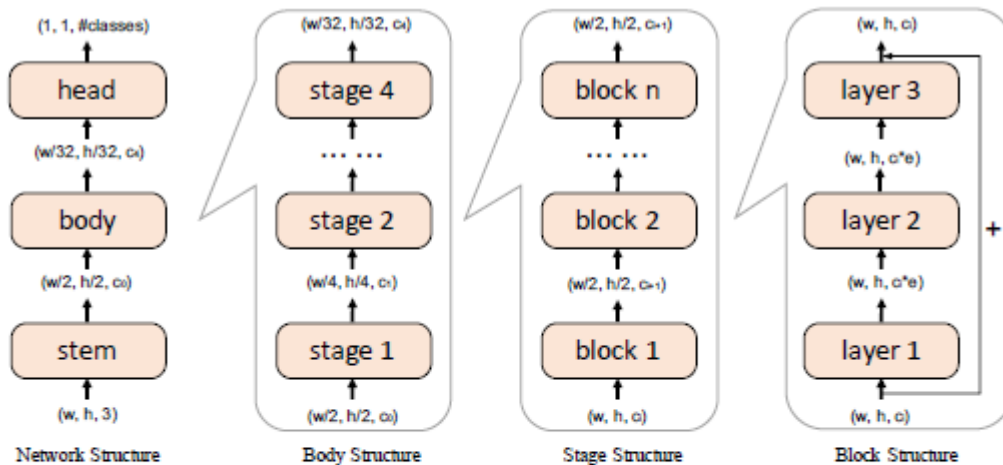
$$\min_{\hat{\mathbf{w}}} \mathbb{E} \left[\Delta \mathbf{z}^{(\ell),T} \mathbf{H}(\mathbf{z}^{(\ell)}) \Delta \mathbf{z}^{(\ell)} \right] = \min_{\hat{\mathbf{w}}} \mathbb{E} \left[\Delta \mathbf{z}^{(\ell),T} \text{diag} \left(\left(\frac{\partial L}{\partial \mathbf{z}_1^{(\ell)}} \right)^2, \dots, \left(\frac{\partial L}{\partial \mathbf{z}_a^{(\ell)}} \right)^2 \right) \Delta \mathbf{z}^{(\ell)} \right]$$

03. Proposed Method

Block Reconstruction

However, if **more layers are considered**, the distance of output will become an inaccurate signal and prohibits the optimization.

Layer가 많아지면 최적화가 부정확해지므로 그부분은 고려해봐야함



(a) A typical structure of CNN (taken from Radosavovic et al. (2020)). Network is composed of a stem layer (first convolution on input images), a body and a head layer (average pooling with a fully connected layer). A body contains several stages, and a stage contains several blocks. A representative block is the bottleneck block with residual path.

$$\bar{\mathbf{g}}^{(\mathbf{w})} = \mathbb{E}[\nabla_{\mathbf{w}} L]$$

$$\bar{\mathbf{H}}^{(\mathbf{w})} = \mathbb{E}[\nabla_{\mathbf{w}}^2 L]$$

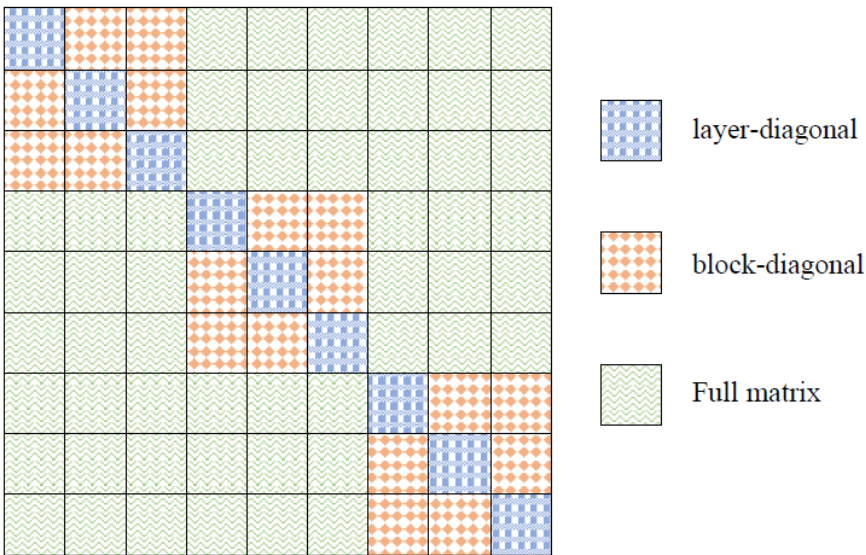
$$\mathbb{E}[L(\mathbf{w} + \Delta \mathbf{w})] - \mathbb{E}[L(\mathbf{w})] \approx \Delta \mathbf{w}^T \bar{\mathbf{g}}^{(\mathbf{w})} + \frac{1}{2} \Delta \mathbf{w}^T \bar{\mathbf{H}}^{(\mathbf{w})} \Delta \mathbf{w}$$

1. Layer-wise Reconstruction
2. Block-wise Reconstruction
3. Stage-wise Reconstruction
4. Network-wise Reconstruction

03. Proposed Method

Block Reconstruction

However, if **more layers are considered**, the distance of **output will become an inaccurate signal** and prohibits the optimization.



(b) An example illustration of Hessian (or Fisher) matrix. Blue sub-block means the layer-diagonal and each layer are mutual-independent; orange sub-block consider the dependency inside a building block and green parts measure all dependencies.

$$\bar{\mathbf{g}}^{(\mathbf{w})} = \mathbb{E}[\nabla_{\mathbf{w}} L]$$

$$\bar{\mathbf{H}}^{(\mathbf{w})} = \mathbb{E}[\nabla_{\mathbf{w}}^2 L]$$

$$\mathbb{E}[L(\mathbf{w} + \Delta \mathbf{w})] - \mathbb{E}[L(\mathbf{w})] \approx \Delta \mathbf{w}^T \bar{\mathbf{g}}^{(\mathbf{w})} + \frac{1}{2} \Delta \mathbf{w}^T \bar{\mathbf{H}}^{(\mathbf{w})} \Delta \mathbf{w}$$

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03. Proposed Method

Block Reconstruction

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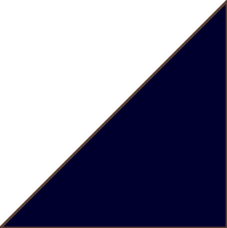
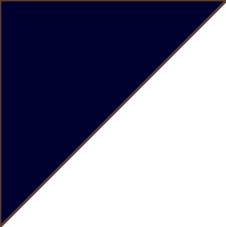
$$\mathbb{E}[L(\mathbf{w} + \Delta \mathbf{w})] - \mathbb{E}[L(\mathbf{w})] \approx \Delta \mathbf{w}^T \bar{\mathbf{g}}^{(\mathbf{w})} + \frac{1}{2} \Delta \mathbf{w}^T \bar{\mathbf{H}}^{(\mathbf{w})} \Delta \mathbf{w}$$

To reach the best tradeoff between them, we use the building **block as the basic reconstruction unit**.

Table 1: Ablation study.

Model	Layer	Block	Stage	Net
ResNet-18	65.19	66.39	66.01	54.15
MobileNetV2	52.13	59.67	54.23	40.76

1. Layer-wise Reconstruction
2. Block-wise Reconstruction
3. Stage-wise Reconstruction
4. Network-wise Reconstruction



Q&A

04. Experiments

Imagenet - PTQ

Table 2: Accuracy comparison on *weight-only quantized* post-training models. Activations here are un-quantized and kept full precision. We also conduct variance study for our experiments. Bold values indicates best results. * indicates our implementation based on open-source codes.

Methods	Bits (W/A)	ResNet-18	ResNet-50	MobileNetV2	RegNet-600MF	RegNet-3.2GF	MnasNet-2.0
Full Prec.	32/32	71.08	77.00	72.49	73.71	78.36	76.68
Bias Correction*	4/32	50.43	64.64	62.82	67.09	71.73	72.31
OMSE (Choukroun et al., 2019)	4/32	67.12	74.67	-	-	-	-
AdaRound (Nagel et al., 2020)	4/32	68.71	75.23	69.78	71.97*	77.12*	74.87*
AdaQuant (Hubara et al., 2020)	4/32	68.82	75.22	44.78	-	-	-
Bit-Split (Wang et al., 2020)	4/32	69.11	75.58	-	-	-	-
BRECQ (Ours)	4/32	70.70$\pm$0.07	76.29$\pm$0.04	71.66$\pm$0.04	73.02$\pm$0.09	78.04$\pm$0.04	76.00$\pm$0.02
Bias Correction*	3/32	12.85	7.97	10.89	28.82	17.95	40.72
AdaRound (Nagel et al., 2020)*	3/32	68.07	73.42	64.33	67.71	72.31	69.33
AdaQuant (Hubara et al., 2020)*	3/32	58.12	67.61	12.56	-	-	-
Bit-Split (Wang et al., 2020)	3/32	66.75	73.24	-	-	-	-
BRECQ (Ours)	3/32	69.81$\pm$0.05	75.61$\pm$0.09	69.50$\pm$0.12	71.48$\pm$0.07	77.22$\pm$0.04	74.58$\pm$0.08
Bias Correction*	2/32	0.13	0.12	0.14	0.18	0.11	0.11
AdaRound (Nagel et al., 2020)*	2/32	55.96	47.95	32.54	25.66	24.70	30.60
AdaQuant (Hubara et al., 2020)*	2/32	0.30	0.49	0.11	-	-	-
BRECQ (Ours)	2/32	66.30$\pm$0.12	72.40$\pm$0.12	59.67$\pm$0.13	65.83$\pm$0.13	73.88$\pm$0.14	67.13$\pm$0.13

04. Experiments

Imagenet - PTQ

Table 3: Accuracy comparison on *fully quantized* post-training models. Activations here are quantized to 4-bit. Notations follows the upper table.

Methods	Bits (W/A)	ResNet-18	ResNet-50	MobileNetV2	RegNet-600MF	RegNet-3.2GF	MNasNet-2.0
Full Prec.	32/32	71.08	77.00	72.49	73.71	78.36	76.68
ACIQ-Mix (Banner et al., 2019)	4/4	67.0	73.8	-	-	-	-
ZeroQ (Cai et al., 2020)*	4/4	21.71	2.94	26.24	28.54	12.24	3.89
LAPQ (Nahshan et al., 2019)	4/4	60.3	70.0	49.7	57.71*	55.89*	65.32*
AdaQuant (Hubara et al., 2020)	4/4	67.5	73.7	34.95*	-	-	-
Bit-Split (Wang et al., 2020)	4/4	67.56	73.71	-	-	-	-
BRECQ (Ours)	4/4	69.60$\pm$0.04	75.05$\pm$0.09	66.57$\pm$0.67	68.33$\pm$0.28	74.21$\pm$0.19	73.56$\pm$0.24
ZeroQ (Cai et al., 2020)*	2/4	0.08	0.08	0.10	0.10	0.05	0.12
LAPQ (Nahshan et al., 2019)*	2/4	0.18	0.14	0.13	0.17	0.12	0.18
AdaQuant (Hubara et al., 2020)*	2/4	0.21	0.12	0.10	-	-	-
BRECQ (Ours)	2/4	64.80$\pm$0.08	70.29$\pm$0.23	53.34$\pm$0.15	59.31$\pm$0.49	67.15$\pm$0.11	63.01$\pm$0.35

04. Experiments

Imagenet - QAT

Table 4: Performance as well as training cost comparison with quantization-aware training (QAT).

Models	Methods	Precision	Accuracy	Model Size	Training Data	GPU hours
ResNet-18 FP: 71.08	ZEROQ (CAI ET AL., 2020))	4/4	21.20	5.81 MB	0	0.008
	BRECQ (OURS)	4/4	69.60	5.81 MB	1024	0.4
	BRECQ (W/ DISTILLED DATA)	4/4	69.32	5.81 MB	0	0.4
	PACT (CHOI ET AL., 2018)	4/4	69.2	5.81 MB	1.2 M	100
	DSQ (GONG ET AL., 2019)	4/4	69.56	5.81 MB	1.2 M	100
	LSQ (ESSER ET AL., 2020)	4/4	71.1	5.81 MB	1.2 M	100
MobileNetV2 FP: 72.49	BRECQ (OURS)	4/4	66.57	2.26 MB	1024	0.8
	PACT (CHOI ET AL., 2018)	4/4	61.40	2.26 MB	1.2 M	192
	DSQ (GONG ET AL., 2019)	4/4	64.80	2.26 MB	1.2 M	192
	BRECQ (OURS)	Mixed/8	70.74	1.38 MB	1024	3.2
	HAQ (WANG ET AL., 2019)	Mixed/8	70.90	1.38 MB	1.2 M	384

04. Experiments

MS COCO

Table 5: Object detection task (MS COCO) comparison on *fully quantized* post-training models. Activations here are quantized to 8-bit. We report the bounding box mean Average Precision (mAP) metric.

Models	Backbone	Full Prec.	Bias Correction*		AdaRound*		ZeroQ	BRECQ (Ours)		
		32/32	8/8	4/8	4/8	2/8	4 _{MP} /8	8/8	4/8	2/8
Faster RCNN (Ren et al., 2015)	ResNet-18	34.55	34.30	0.84	33.96	23.01	-	34.53	34.34	31.82
	ResNet-50	38.55	38.25	0.25	37.58	19.63	-	38.54	38.29	34.23
	MobileNetV2	33.44	33.24	18.39	32.77	16.35	-	33.40	33.18	27.54
RetinaNet (Lin et al., 2017)	ResNet-18	33.20	33.00	0.04	32.59	19.93	-	33.14	33.01	31.42
	ResNet-50	36.82	36.68	0.07	36.00	19.97	33.7	36.73	36.65	34.75
	MobileNetV2	32.63	32.60	18.47	31.89	14.10	-	32.57	32.31	27.59



Thank you